Computer Applications in Business

Lecture 2
Looking Inside the computer System

Farjana Rahman Farin
Lecturer, Dept. of CSE, Sonargaon University

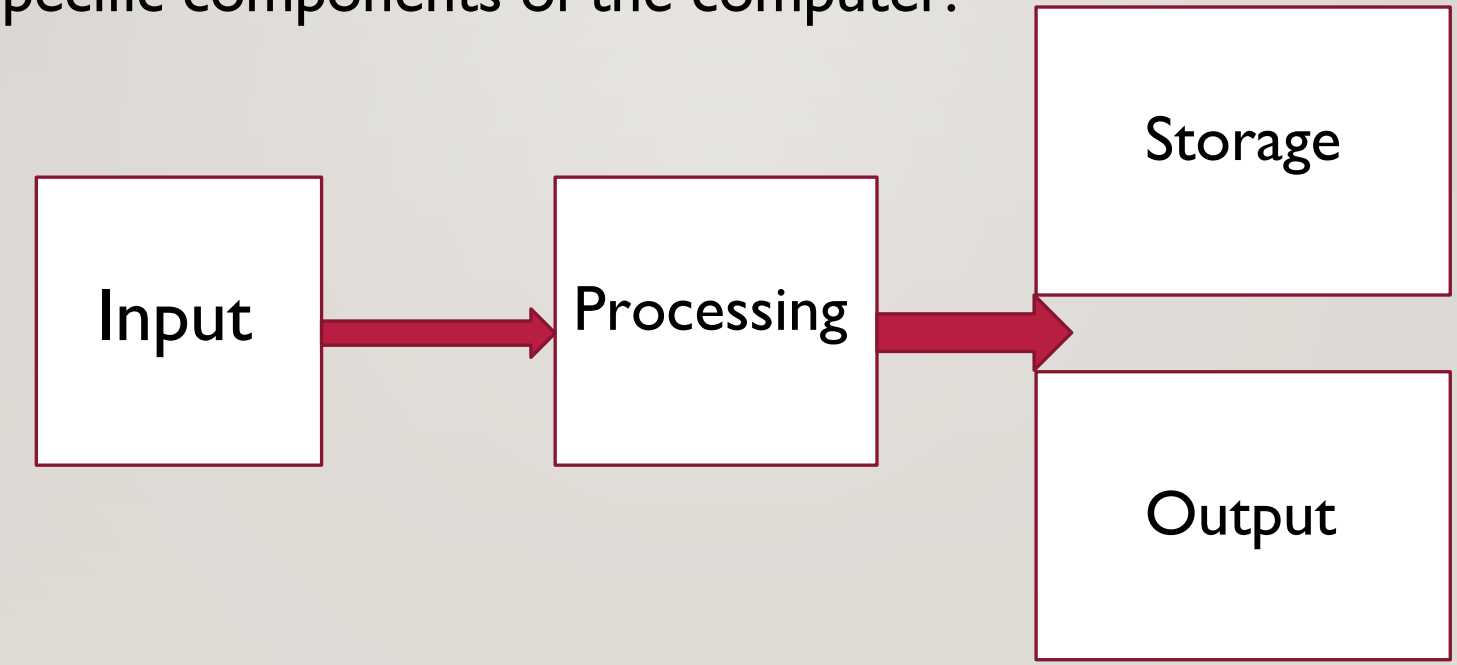
A complete computer system consists of four parts

- ✔ Hardware: The mechanical devices that make up the computer are called hardware. Hardware is any part of the computer you can touch.
- ✔ Software: Software is a set of instructions that makes the computer perform tasks.
- ✔ Data: Data consist of individual facts or pieces of information that by themselves may not make much sense to a person
- ✔ User: People are the computer operators, also known as users

The Information Processing Cycle:

Computer takes data, then processes the data as per being instructed and then can either output the generated information to user or store the information to storage or can do both.

The information processing cycle has four parts, and each part involves one or more specific components of the computer:



Essential Computer Hardware

A computer's hardware devices fall into one of four categories:

1. Processor
2. Memory
3. Input and output
4. Storage

While any type of computer system contains these four types of hardware.

Processor Devices: The procedure that transforms raw data into useful information is called processing. To perform this transformation, the computer uses two components:

- the processor
- memory.

processor is like the brain of the computer; it organizes and carries out instructions that come from either the user or the software

Memory Devices

Memory: memory is one or more sets of chips that store data and/or program instructions, either temporarily or permanently.

- Personal computers use several different types of memory, but the two most important arc called random access memory (RAM) and read-only memory (ROM). These two types of memory work in very different ways and perform distinct functions.

Storage Devices: The purpose of storage is to hold data permanently, even when the computer is turned off.

HW

Computer memory is broadly classified into three categories – **Primary Memory, Secondary Memory,** and **Cache Memory.**

Primary memory	Secondary memory
Primary memory is one that stores data and instructions which are currently being processed by the system or by the CPU.	**Secondary memory** is one that used to store data, programs and information permanently on the computer system.
Primary memory is called the internal memory or main memory.	It is an external memory of the computer .It is also called auxiliary memory or permanent memory.
It is a volatile memory, because when the power is switched off, it loses all its data.	It is a type of non-volatile memory because it retains the data even if the power is switched off.
Primary memory is faster than secondary memory because it is directly accessible to the CPU.	Secondary memory is less faster than primary memory.
The primary memory is further classified into two types namely, **RAM** and **ROM.**	Examples of secondary memories include hard disc, solid state drive, CD, DVD, pen drive, blue ray disc, SD-Card, etc.

Input and Output Devices: Input devices accept data and instructions from the user or from another computer system (such as a computer on the Internet). Output devices return processed data to the user or to another computer system.

- Input Devices:

- **Keyboard :** The keyboard is an essential input device and computer and laptops both use keyboards to give commands to the computer.

- **Mouse:**
  - It is also known as a pointing device.
  - A mouse comprises 3 buttons on the top and one trackball at the bottom which helps in selecting and moving the mouse around, respectively

- **Light Pen:** It is a wand-like looking device which can directly be moved over the device's screen. It is light-sensitive.

- **Microphone:** Using a microphone, sound can be stored in a device in its digital form. It converts sound into an electrical signal.

- **Scanner:** This device can scan images or text and convert it into a digital signal. When we place any piece of a document on a scanner, it converts it into a digital signal and displays it on the computer screen

Input Devices

KEYBOARD

MOUSE

SCANNER

LIGHT PEN

WEB CAMERA

JOYSTICK

TRACK BALL

Output Devices:

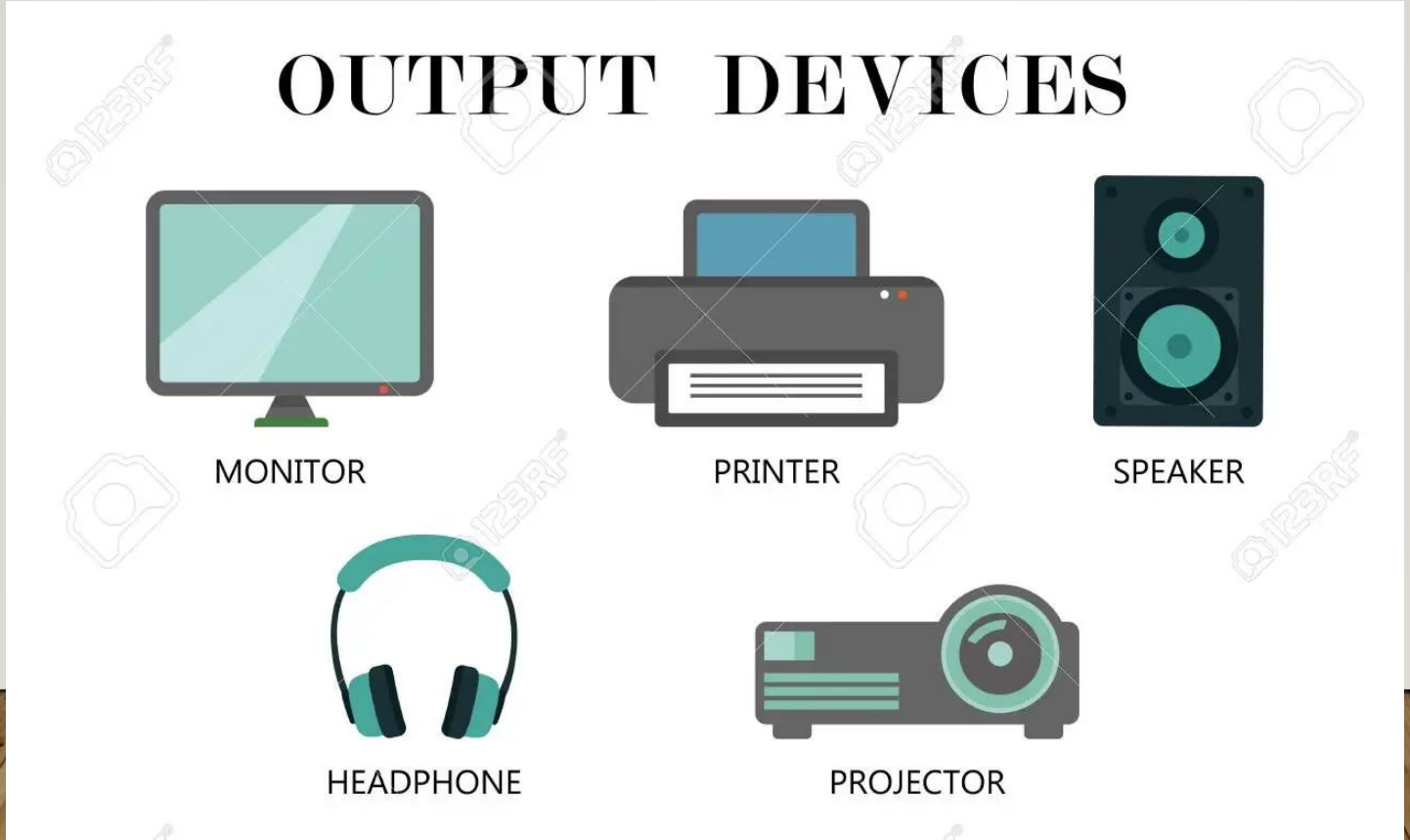
- **Monitor:** The device which displays all the icons, text, images, etc. over a screen is called the Monitor.

- **Printer:** A device which makes a copy of the pictorial or textual content, usually over a paper is called a printer.

- **Speakers:** A device through which we can listen to a sound as an outcome of what we command a computer to do is called a speaker.

- **Projector:** An optical device which presents an image or moving images onto a projection screen is called a projector.

- **Headphones:** They perform the same function as a speaker, the only difference is the frequency of sound. Using speakers, the sound can be heard over a larger area and using headphones, the sound is only audible to the person using them



Software Brings the Machine to Life

The ingredient that enables a computer to perform a specific task is software, which consists of instructions. A set of instructions that drive a computer to perform specific tasks is called a program. These instructions tell the machine's physical components what to do; without the instructions, a computer could not do anything at all.

System Software:

System software is any program that controls the computer's hardware or that can be used to maintain the computer in some way so that it runs more efficiently. There are three basic types of system software:

1. An operating system tells the computer how to use its own components. Examples of operating systems include Windows and Linux. When a program wants the hardware to do something, it communicates through the operating system.

1. A network operating system allows computers to communicate and share data across a network while controlling network operations and overseeing the network's security.
2. A utility is a program that makes the computer system easier to use or performs highly specialized functions.

Application Software:

Application software tells the computer how to accomplish specific tasks, such as word processing or drawing, for the user. Some of the major categories of these applications include.

- Word processing software for creating text-based documents such as newsletters or brochures.
- Spreadsheets for creating numeric-based documents such as budgets or balance sheets.
- Database management software for building and manipulating large sets of data, such as the names, addresses, and phone numbers in a telephone directory.

Computer Use & Users Role

When working with a personal computer; the user can take on several roles, depending on what he or she wants to accomplish:

- Setting up the System.
- Installing Software.
- Running Programs.
- Managing Files
- Maintaining the System